SET-1

**Roll No.** | | | | | | | |

# END SEMESTER EXAMINATION 2025

*Name of the Course*: B.Tech Petroleum Engineering — *Semester: VI*

*Name of the Paper*: Reservoir Modeling & Simulation — *Course Code:* TPE-601

Time: Three Hours — MM: 100

**Note:**

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

| Q1 | | COs |
|---|---|---|
| a. | What are the three essential properties of a Reservoir rocks from hydrocarbon point of view? Explain each in detail. Name the various type of physical properties of rock? | CO-1 |
| b. | What is the different type of sedimentary rocks are available on the earth? Explain their composition and relative abundance on the earth. | CO-2 |
| c. | What is Surface & Interfacial tension explain with diagram? What is wetting & non wetting phase of a reservoir? | CO-2 |
| Q2 | | |
| a. | What do you understand by petroleum system? Draw a cross section of a petroleum system and explain its all-important components. | CO-2 |
| b. | Construct a diagram showing Generation, Migration and trapping of hydrocarbons of a Petroleum system. | CO-2 |
| c. | Explain Reservoir Modeling, Reservoir Engineering and various step of Reservoir Engineering | CO-3 |
| Q3 | | |
| a. | Explain drive mechanism of petroleum system. Explain Primary and secondary recovery mechanism. | CO-3 |
| b. | Explain natural and artificial recovery Mechanisms in detail. How many types of recovery mechanism are available in oil industry? | CO-4 |
| c. | Explain Reservoir pressure, Rock texture, Porosity, Permeability and saturation of a reservoir rock. | CO-4 |
| Q4 | | |
| a. | Explain Darcy equation, how it is used to measure permeability for flowing fluid and gas system? | CO-5 |
| b. | Draw a sequential reservoir modeling workflows for building a reservoir model. Explain its each component. | CO-5 |

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|---|---|---|---|
| | c. | Explain Porosity, Total & Effective Porosity, Permeability, and oil saturation<br>Define with figure Cubical packing, orthorhombic-packing & rhombohedra – packing of spheres | CO-4<br>CO-5 |
| **Q5** | | | |
| | a. | Construct a suitable workflow for fault and structure model building of a reservoir model in RMS. | CO-5 |
| | b. | What is facies modeling? Design a workflow for Facies modeling. Explain its all components. Why it is required? | CO-5 |
| | c. | Categorize and explain in detail the various types of data required for facies log creation, fault modeling, structure & Property model | CO-5 |